

Pranav Karthik A

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ABOUT

Detail-oriented Electronics and Communication Engineering student with a solid foundation in Data Science and Programming. Currently experimenting with integrating machine learning and signal processing , application development and IoT system integration. Passionate about bridging the gap between hardware systems and scalable software solutions.

SKILLS

Languages: Python, C++, Java, SQL , Verilog, Matlab.

Tools: MATLAB/Simulink, LTspice, GitHub, VS Code, IntelliJ IDEA, Google Colab, Jupyter Notebook, Android Studio, Figma, Canva.

EDUCATION

Sri Sivasubramaniya Nadar College of Engineering

Chennai, India

B.E. in Electronics and Communication Engineering (Semester IV)

2024 – 2028

- **Grade:** 8.69 / 10.0 CGPA (Current)
- **Relevant Coursework:** Linear Algebra, Signals and Systems, Digital Signal Processing, Object Oriented Programming, Data Structures, Digital System Design.

Indian Institute of Technology, Madras

Chennai, India

BS in Data Science and Applications (Diploma Level - Programming)

2024 – 2028

- **Grade:** 9.5 / 10.0 CGPA (Current)
- **Relevant Coursework:** Programming and Data Structures and Algorithms in Python (PDSA), Programming using Java, Database Management Systems (DBMS), Statistics and Mathematics for Data Science.

EXPERIENCE

Machine Learning Intern

Jul. 2025 – Aug. 2025

LaunchEd Global

Chennai, India

- Developed a predictive model using **Linear Regression** to estimate candidate CTC, identifying key salary drivers from a dataset of **25,000+ records**.
- Performed end-to-end **Exploratory Data Analysis (EDA)** and data preprocessing, including feature engineering and handling missing values to improve model reliability.
- Implemented and compared **Regression models and Decision Trees** using Python, ensuring unbiased estimation and high predictive accuracy on real-world employee data.
- Automated the algorithmic pipeline for data cleaning, reducing manual processing time and ensuring consistency across diverse data features.

PROJECTS

AI-Powered Transit Tracker (SSNbustrackerapp)

- * Engineered a robust Android application integrating **Firestore Realtime Database** and **OSMDroid** for live location tracking and dynamic geospatial visualization.
- * Deployed a custom **Gemini API** integration to process live GPS coordinates and generate natural language status updates, providing context-aware arrival predictions.
- * Optimized backend synchronization and **API polling logic** to maintain low-latency updates while minimizing battery consumption and data overhead.

Predictive Salary Modeling (CTC Estimation)

- * Architected a high-precision predictive model using **Linear Regression** to estimate candidate CTC based on a dataset of **25,000+ records**.
- * Conducted extensive **Exploratory Data Analysis (EDA)** and feature engineering, implementing statistical imputation to significantly reduce Mean Squared Error.
- * Utilized **Python (Pandas, Scikit-Learn, Matplotlib)** to visualize correlation matrices and feature importance across diverse professional demographics.

Smart Shadow Detection System

- * Designed and prototyped an analog detection system utilizing **photodiodes** and a multi-stage **Darlington amplifier** circuit to transform light variations into measurable voltage signals.
- * Implemented hardware-level **signal conditioning** and noise filtration to enhance sensor sensitivity for reliable detection across varying ambient light environments.
- * Validated circuit performance through simulation to ensure immediate response times for real-time light-sensing applications.

Verilog-Based Electronic Voting Machine (EVM)

- * Designed and simulated a digital Electronic Voting Machine using **Verilog HDL**, incorporating asynchronous counters and shift registers to manage and store real-time vote tallies.
- * Implemented custom control logic for candidate selection and secure reset functionality, validating the architecture through comprehensive **Testbench** simulations.
- * Optimized the register-transfer level (RTL) design to ensure synchronized data flow, successfully handling multi-candidate inputs without data race conditions.